

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

1. Amount of Data Rendered to Database

The project processes and stores large volumes of electricity consumption data collected from various sources such as smart meters, utility providers, and regional energy departments. The database contains historical and real-time records, enabling comprehensive analysis of energy usage trends. Efficient database management ensures quick retrieval of information and supports large-scale analytical operations.

Key Highlights:

- Stores consumer electricity usage records.
- Supports historical and real-time data analysis.
- Handles thousands to millions of data entries.
- Enables efficient querying and reporting.

2. Utilization of Data Filters

Data filters play a crucial role in narrowing down large datasets and extracting meaningful insights. Users can apply various filters to analyze electricity consumption patterns across different dimensions such as time, location, consumer category, and usage levels. These filters improve user experience by allowing customized data exploration and focused decision-making.

Common Filters Used:

- Year and Month
- Geographic Region/State
- Residential, Commercial, and Industrial Consumers
- Peak and Off-Peak Hours
- Consumption Range
- Seasonal Analysis

Benefits:

- Faster data exploration.
- Enhanced decision-making.
- Improved dashboard interactivity.
- Personalized analysis based on user requirements.

3. Number of Visualizations and Graphs

Visualizations transform complex electricity consumption data into easy-to-understand graphical representations. Multiple charts and graphs are used to identify trends, compare usage patterns, and forecast future energy demands. These visual tools help stakeholders quickly interpret data and make informed decisions.

Types of Visualizations Used:

- Line Charts: Display electricity consumption trends over time.
- Bar Charts: Compare energy usage across regions or sectors.
- Pie Charts: Show percentage distribution of consumption categories.
- Area Charts: Illustrate cumulative energy usage patterns.
- Heat Maps: Highlight high and low consumption zones.
- Scatter Plots: Analyze relationships between variables.
- Forecasting Graphs: Predict future electricity demand.

Advantages:

- Simplifies complex datasets.
- Reveals hidden patterns and trends.
- Supports data-driven planning.
- Enhances report presentation and dashboard effectiveness.

4. Insights and Outcomes

The analysis of electricity consumption patterns provides valuable insights into energy demand, peak usage periods, and consumer behaviour. These findings help utility providers optimize energy distribution, reduce wastage, and develop sustainable energy management strategies.

Expected Outcomes:

- Identification of peak consumption periods.
- Better demand forecasting.
- Improved energy efficiency planning.
- Support for smart grid implementation.
- Enhanced sustainability and resource management.

DELTA
— ACHIEVERS —